

Sprint 4

Work summary

Problem Statement & Why It Matters

Frame the scheduling pain points, the Sprint 4 objective, and stakeholder value

Current Limitations

- Current scheduling requires manual regeneration and re-fetching schedules, with no AI assistance today.
- Single-output regeneration makes trade-off comparison (coverage vs. fairness vs. satisfaction) difficult.

Desired Outcome

- Sprint 4 objective: enable AI-powered rescheduling with multiple candidates for planner selection (planner-only scope).
- Stakeholder value: better scheduling quality, retention outcomes, and safety/auditability.

AI provides multiple candidate schedules; planners retain final control.



Baseline Scheduling Flow (Today)

Show the existing generation/regeneration pipeline to anchor the AI extension.

- REST endpoints **/schedule/generation** and **/schedule/regeneration/id={id}** trigger the pipeline; response is status only.
- Orchestrator collects data, builds ConcreteShift range, configures priorities, then runs **ScheduleBuilder.build()**.
- Data sources include shifts, constraints, doctors, holidays, priorities, and snapshots.
- Error handling maps invalid schedules to **406 NOT_ACCEPTABLE**, a pattern reused for AI partials later.

This baseline remains the fallback in AI workflows.

AI Protocol (TOON/JSON) & Data Minimization

Explain how AI is integrated safely and with GDPR minimization.

Direct Prompting

Direct prompting only: system sends TOON context; AI returns strict JSON response.

TOON Input

TOON input is token-efficient and includes schedule window, shifts, doctors, constraints, blocks, and feedbacks.

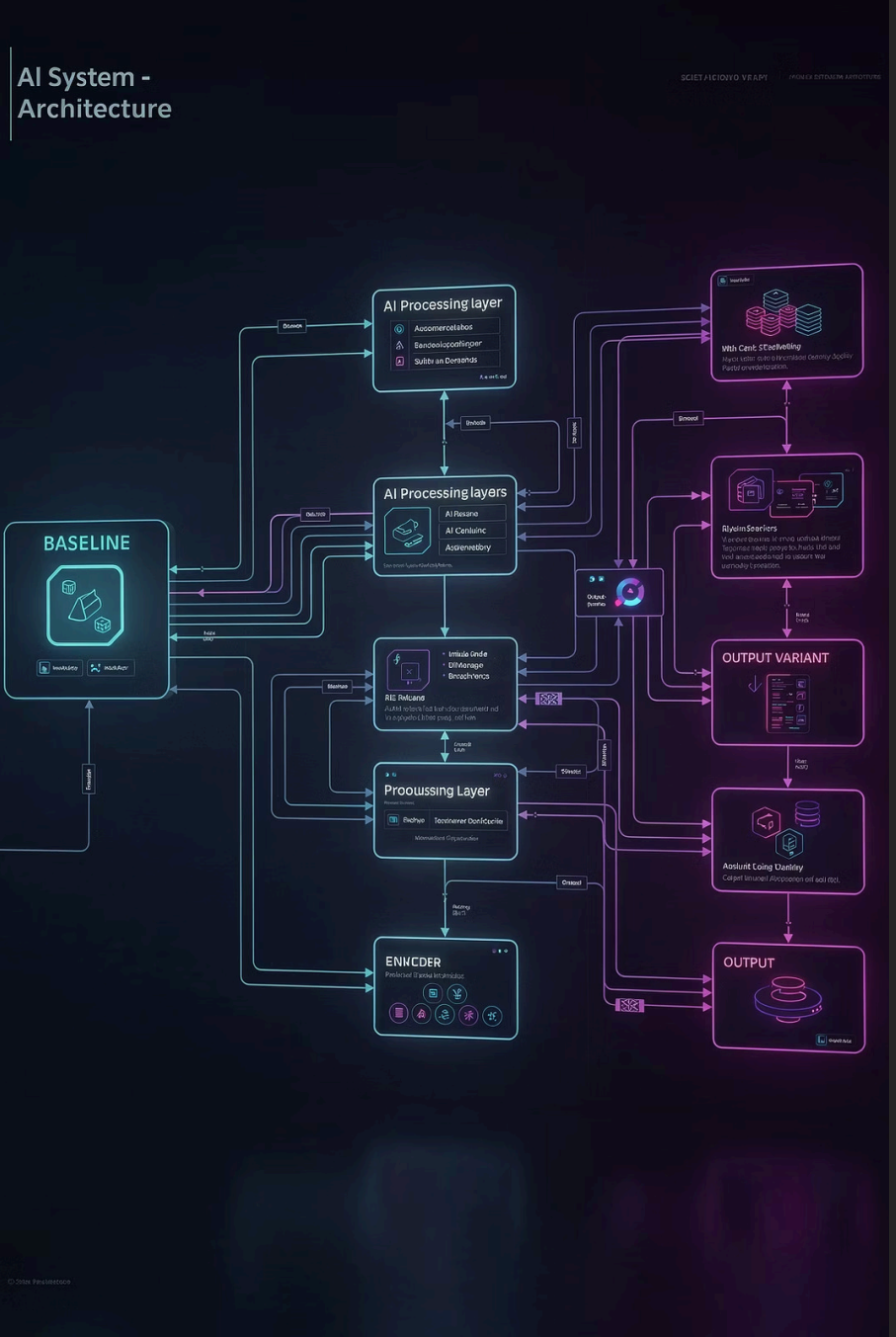
JSON Validation

JSON output must validate schema/semantics; only **SUCCESS** is accepted.

GDPR Compliance

GDPR: pseudonymized doctor IDs, limited feedback context, no direct identifiers.

Protocol is provider-agnostic and privacy-minimized.



Orchestration Pipeline (Baseline + 3 AI Variants)

Show the end-to-end orchestration and selection rule.

- Pipeline generates baseline first, then a single AI call returns three variants: Empatica, Efficiente, Bilanciata.
- Steps: build TOON context → validate → call AI broker → parse JSON → compute metrics → comparison payload.
- Status handling aligns to UI: **success**, **partial**, **failure** with baseline fallback on AI errors.
- Persistence rule: only the planner-selected schedule is saved.

One AI call, three candidate schedules, planner chooses.

Metrics & Decision Framework (GQM-Aligned)

Explain how schedules are compared and ranked.

GQM Goals

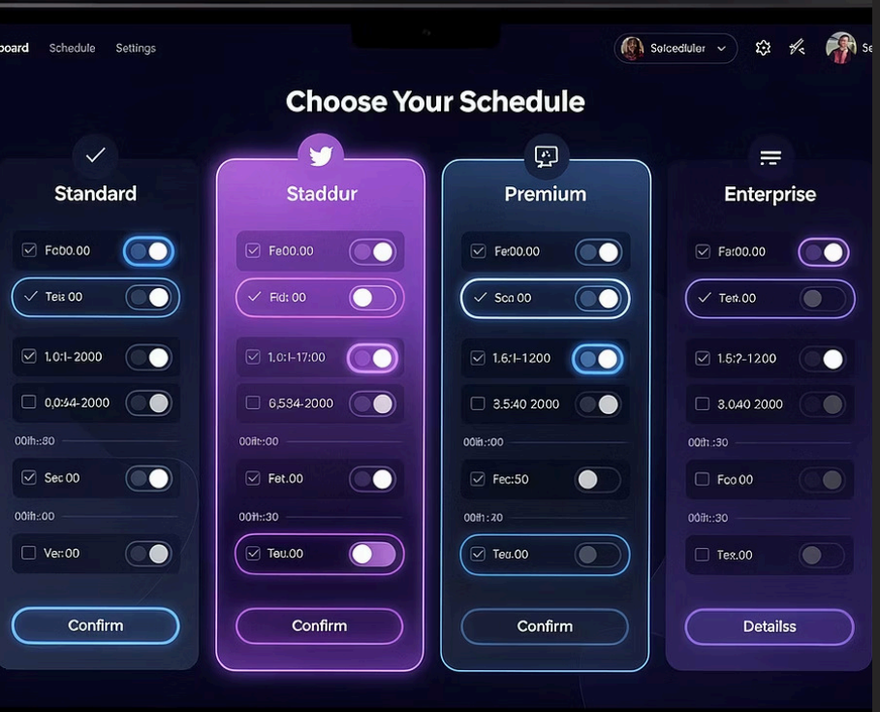
- Reduce burnout/turnover
- Improve satisfaction
- Generate 3 valid AI schedules

Decision Metrics

Decision metrics include coverage, UFFA balance, sentiment transitions, UP delta, variance delta (normalized to [0,1]).

Weighted priority scale drives a deterministic score; tie-break prioritizes coverage first.

Coverage is the primary safety constraint.



UI Comparison Preview & Next Steps

Show planner-facing UI direction + sprint close-out actions.

01

UI Target

UI target: 2×2 comparison grid for baseline + 3 AI candidates with decision metrics only.

02

Selection Flow

Selection flow: planner selects candidate, confirms in modal, choice is locked.

03

Feedback

Success/error feedback reuses existing UI patterns.

04

Next Steps

Next steps: finalize comparison layout + selection wiring; align with backend comparison schema.

UI readiness is the key adoption blocker; prioritize completion.



Grazie dell'attenzione

CoffeeBreakpoint